

Laminated and Heterogeneous Integration for Opto-electronics via Nano-printing

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Abstract

We established a nano-resolution printing method to deposit nanocrystal (NC) inks and modify material properties via in-situ room-temperature ligand-exchange. This enables heterogeneous integration of multi-layer opto-electronic devices on low-thermal-resistance substrates. We printed metal (Au, Ag) and semiconductors (ZnO, PbS, CdSe, and InSb) NC structures, achieving 70-nm linewidths with competitive (opto)electronic characteristics for micro-nano devices.

Author Keywords

nano-printing; heterogeneous integration; room temperature; ligand-exchange; low-thermal-resistance substrate.

1. Introduction

The exponential proliferation of the Internet of Things (IoT), wearable technology, autonomous systems, and augmented and virtual reality (AR/VR) has rendered ubiquitous sensing and computing a reality, thereby precipitating a demand for high-performance, application-specific optoelectronics.^{1,2} The realization of such customizable devices necessitates engineered multi-material constituents, adaptable manufacturing methodologies, and flexible heterogeneous integration strategies. While additive manufacturing facilitates the fabrication of bespoke, multiscale structures via functional inks, current techniques remain constrained. For instance, although stereolithography attains sub-micrometer resolution, its material palette is primarily restricted to metal oxides and polymers, limiting its utility largely to mechanical and optical components.^{3,4} Consequently, fully printed, pixelated microelectronics—which require precise semiconductor patterning and multilayer stacking—remain at a nascent stage of development. Addressing these challenges necessitates an adjustable ink system integrated with versatile printing protocols. Colloidal nanocrystals (NCs) are promising candidates, having demonstrated efficacy in flexible circuits and photodetectors. Their crystalline core-shell structure offers tailorable properties, particularly the size-tunable optoelectronics of NCs.^{5,6} Crucially, ligand-exchange (L-E) substitutes insulating ligands with compact alternatives to enable charge transport.⁷ Moreover, selective ligand chemistry allows for the engineering of carrier mobility, doping, and energy levels. This facilitates the layer-by-layer assembly of complex devices while mitigating micro-defects associated with ligand loss, thereby supporting robust multi-material integration.

Electrohydrodynamic printing (EHDP) enables nanoscale resolution with diverse ink materials, overcoming nozzle diameter limitations.⁸ However, conventional high-temperature sintering is often deleterious to heat-sensitive semiconductive NCs and flexible substrates.^{9,10} While low-temperature chemical treatments have been used to deposit NCs as sensitizers on graphene¹¹ for infrared detection, achieving scalable all-printed NC optoelectronics requires deeper understanding of NC-surface ligand interactions and multi-material integration.

We present a ligand-exchange-assisted EHDP protocol enabling

sub-micron NC patterning with in-situ room-temperature functionalization. This method facilitates the heterogeneous integration of Ag, Au, ZnO, and PbS NCs, achieving linewidths of 70 nm and Ag conductivities of 1.99×10^7 S/m via post-exchange densification. Beyond fabricating structural color gratings and metasurfaces, we demonstrate multi-layer IR photodiodes featuring 10- μ m pixels composed of n- and p-doped PbS NCs with asymmetric contacts. Under 1500 nm illumination, these devices exhibit enhanced rectification, reduced dark currents, and rapid response times compared to standard benchmarks. This work establishes a viable room-temperature pathway for processing sensitive materials in advanced optoelectronics.

2. Mechanisms of L-E assisted nano-printing of NC assemblies

Colloidal NCs are typically synthesized with long organic capping ligands, which render them stable as dispersions in non-polar solvents—a critical prerequisite for formulating printable inks compatible with EHDP. As illustrated in **Fig. 2a**, a diverse library of NC core materials can be accessed through this approach. This library encompasses metals, semiconductors (including quantum dots), and metal oxides, all synthesized with surface ligands such as oleic acid (OA) or oleylamine (OLAM). These ligands confer solubility and colloidal stability in non-polar solvent systems like dodecane, thereby enabling the controlled deposition required for high-resolution additive manufacturing.¹² They are deposited by EHDP similar with previous research,⁸ which utilizes pulsed voltages to generate electric forces that overcome ink surface tension and viscosity, enabling droplet ejection from the nozzle tip. As illustrated in **Fig. 1**, by translating the stage of the custom-built EHD printer along a defined path, NCs are patterned into linear structures on the substrate (**Fig. 2b**). These printed lines serve as fundamental building blocks that can be spatially organized and combined through programmed stage movements to form customized patterns and arbitrarily shaped thin films.

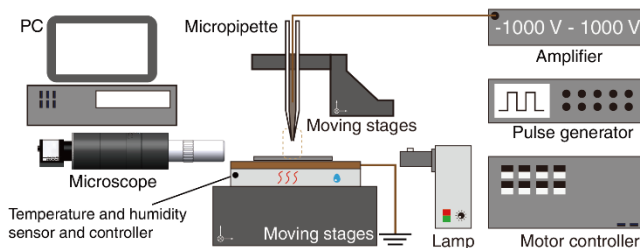


Figure 1. Schematic of the nanoscale electrohydrodynamic (EHD) printer.

As-printed NCs structures (**Fig. 2b**) are initially electrically insulated due to steric separation by bulky organic ligands in the original inks (**Fig. 2a**). To address this, we employ a solid-state L-E process wherein structures are treated with compact ligands (EDT, NH₄SCN, TBAI, or PbI₂) in polar solvents for 30–120 s, followed by rinsing (**Fig. 2c, e**). These thermodynamically

spontaneous room-temperature reactions replace long-chain molecules, thereby reducing interparticle spacing. Consequently, ligand volume loss and altered surface energy drive autonomous NC reorganization, densification, and partial fusion, yielding significant morphological and physical transformations (**Fig. 2d**).

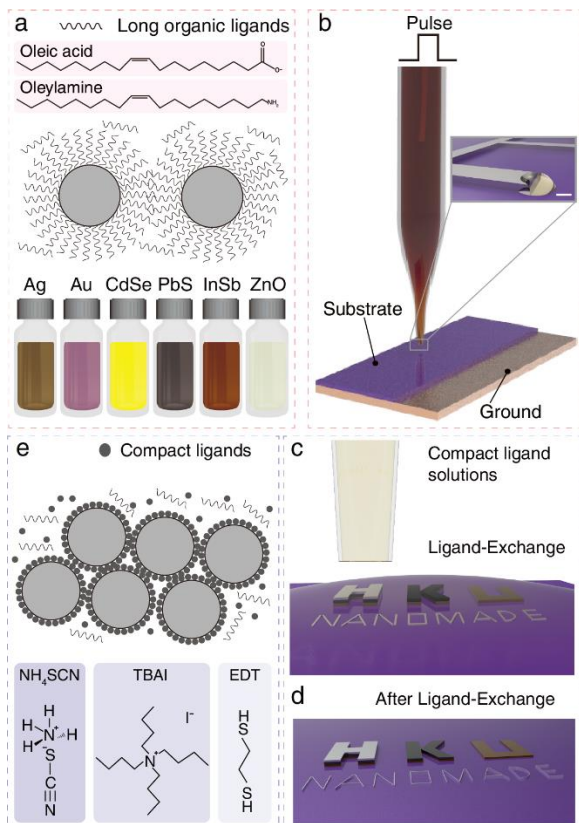


Figure 2. L-E-assisted nanoprinting of NCs. **a** Schematic representation of as-synthesized nanocrystals capped with long organic ligands and their dispersion as stable, printable inks in non-polar solvents. **b** Illustration of the EHDP process for the precise deposition of NC inks onto a substrate. Schematics of **c** the L-E process and **d** morphological changes induced by L-E in patterns produced via multi-layer and multi-material printing. **e** Schematics showing neighboring NCs after L-E treatment with various compact ligand reagents.

Applying a L-E condition of 1 mg/mL NH_4SCN or TBAI for 60 s induces significant volumetric shrinkage in printed structures. Specifically, NH_4SCN treatment reduced Ag NC linewidths from 651.4 ± 49 nm to 274.5 ± 31 nm, while TBAI exchange PbS NC lines from 754.9 ± 65 nm to 544.0 ± 34 nm. This room-temperature assembly yields dense, electronically coupled sub-micron solids with optoelectronic properties comparable to spin-coated benchmarks. The resulting Ag and PbS NC lines and films demonstrate optoelectronic properties on par with their spin-coated counterparts, the underlying mechanisms and optimization of which will be detailed in subsequent sections.

3. Exploration of Resolution Limits and Applications of Printed Lines

To optimize morphology, we utilized Ag NC inks with NH_4SCN ligand exchange as a model system to systematically investigate how printing parameters—specifically stage velocity (v) and pulse voltage (V)—govern the morphology of printed lines, using a fixed

nozzle diameter of 2 μm . Lightly doped silicon with native SiO_2 was selected as the substrate to mitigate artifacts associated with conductive auto-focusing and electrostatic repulsion. A series of Ag NC line arrays were printed under varied v and V conditions, and their morphologies and elemental composition were analyzed using SEM and EDX to quantify deposition per unit length.

Results indicate that linewidth decreases with increasing stage speed, as fewer ink droplets and correspondingly less material are deposited per unit length. At speeds exceeding 6.0 $\mu\text{m/s}$, the linewidth of as-printed Ag NCs becomes less sensitive to further increases in v , and the L-E induced contraction is reduced due to insufficient NC mass available for migration and fusion. As illustrated in **Fig. 3**, continuous Ag NC lines with linewidths as narrow as 70 nm are achieved through this L-E-assisted nano-printing approach.

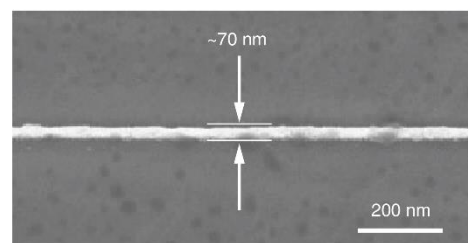


Figure 3. SEM images of Ag NC lines before and after L-E fabricated by EHDP, with a minimum line width of 70 nm.

We draw on the exceptional nanoscale patterning capability and high electrical conductivity of printed silver nanocrystal (Ag NC) lines to create diffraction gratings that produce structural colors (SCs). These gratings spatially decompose incident white light into specific monochromatic components, enabling the generation of diverse color effects. This technology shows considerable potential for applications such as flexible displays, strain sensors, and anti-counterfeiting features.^{13,14} The structural color of the printed Ag NC gratings was precisely modulated by varying the interline spacing from 600 nm to 1900 nm in 100 nm increments, enabling the generation of high-purity structural colors, as illustrated in **Fig. 4**.

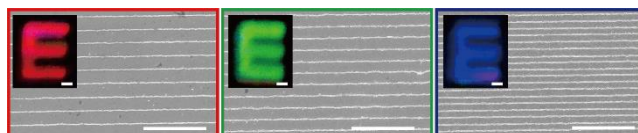


Figure 4. Optical microscopy (top) and corresponding SEM (bottom) images of printed RGB SCs structural color patterns forming the letter 'E', with grating pitch sizes of 2220 nm (left), 1720 nm (middle), and 1320 nm (right), respectively. Scale bars: 10 μm .

By integrating different grating pitches within a single structure, a broader palette of structural colors can be achieved, with the resulting hue determined by the proportional combination of red, green, and blue (RGB) primary color gratings (**Fig. 5**). For instance, an equal-area fraction of the three primary gratings yields a distinct white coloration. In comparison to recently reported meniscus-guided printed polymeric structural colors¹⁴, our approach achieves a significantly wider color gamut (**Fig. 5**). This enhanced performance is attributed to the finer linewidth and greater tunability of line spacing afforded by our method, all while maintaining compatibility with high-speed printing at 25 $\mu\text{m/s}$.

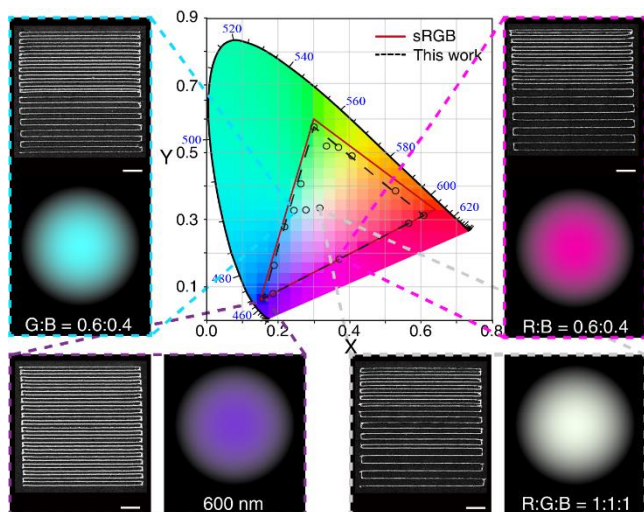


Figure 5. Chromaticity coordinates of representative structural colors from this work plotted on the CIE 1931 chromaticity diagram with SEM images of printed grating structures and their corresponding SCs, demonstrating color tuning achieved through precise control of grating pitch and through mixing of red, green, and blue (RGB) primary SC gratings at varying ratios. Scale bars: 10 μm .

Besides line gratings, we print Ag NCs for complex linear patterns with extended functionalities, including bio-electrode arrays and long-wavelength IR metasurfaces.

4. Optimization of Printed Pattern Conductivity

Upon chemical treatment of nano-printed NC structures, we observe that grain growth induced by L-E is governed more decisively by the initial ink concentration used during the EHDP process. This concentration directly influences the dynamic detachment of ligands from NC surfaces. As described by the Allen-Cahn equation, the remaining surface-bound ligands establish the driving force for subsequent grain growth.¹² Integrating experimental data with computational modeling reveals that a substantial fraction of organic ligands detach during EHDP when higher-concentration NC inks are employed. This pre-deposition ligand loss leads to the formation of larger NC aggregates upon printing, consequently reducing the driving force for chemically driven sintering during the subsequent L-E step. By systematically optimizing the initial ink concentration, we successfully minimized ligand detachment during printing, thereby preserving a higher effective NC concentration. This optimization not only enhances printing efficiency but also markedly improves the filling density and structural integrity of the resulting nanocrystal architectures.

The electrical performance of these structures was evaluated by measuring the DC conductivity of Ag NC films using the four-point probe method. Peak conductivity, reaching $1.99 \times 10^7 \text{ S/m}$, was achieved at an optimized ink concentration of 3.0 mg/mL.

5. Multi-material Laminated Printing for Opto-electronics Fabrication and Heterogeneous Integration

Beyond Ag NCs as printing system, the ligand-exchange (L-E)-assisted nanoprinting platform demonstrates broad material versatility. Its applicability extends to a range of functional colloidal inks, including semiconductive PbS NCs, InSb NCs, CdSe NCs and dielectric ZnO NCs, showcased with the NH_4SCN L-E to realise various pattern designs at sub-micron resolution,

promoting multi-material printing capabilities, as shown in Fig. 6.

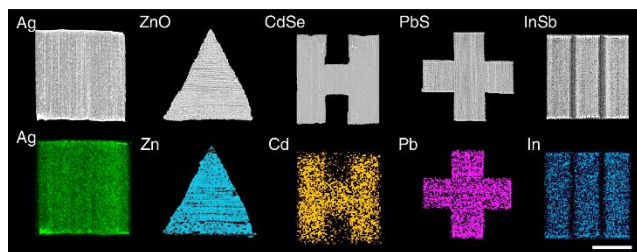


Figure 6. SEM images (top) and corresponding EDX elemental mappings (bottom) of printed NC patterns featuring diverse material compositions and geometries: Ag, ZnO, CdSe, PbS, and InSb NCs. Scale bar: 20 μm .

Leveraging our L-E-assisted nanoprinting platform, we fabricated sub-10 μm PbS NC patterns and integrated photodiode (PD) devices entirely under ambient conditions. In contrast to single-doped PbS NC photoconductors, the photodiodes reported here incorporate built-in P-N junctions. These junctions are realized through the distinct doping characteristics imparted by different ligand treatments: EDT induces p-type doping in PbS NCs, while TBAI or PbI_2 yields n-type behavior.^{15,16} To construct the multi-layer, multi-material device architecture, we first printed bottom electrode pairs using Ag and Au NC inks—each processed with NH_4SCN -assisted ligand exchange. Subsequently, a layer of TBAI-treated n-type PbS NCs was printed over the Ag electrode, followed by a layer of EDT-treated p-type PbS NCs printed to establish direct contact with the Au electrode, as illustrated schematically in Fig. 7.

Stacked PbS NC Photodiodes

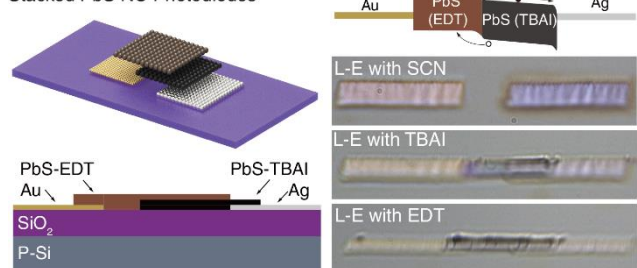


Figure 7. Schematics of all-printed PbS NC IR photodiodes (left), energy band diagram (right top), optical microscopy images (right bottom).

To realize a fully printed nanocrystal photodiode, we incorporated ZnO NC as the n-type electron transport layer. The device architecture consists of multiple NCs layers laterally bridging a pair of pre-printed Au and Ag electrodes, both of which have undergone SCN^- -based ligand exchange. Fabrication proceeds through sequential deposition of PbS-EDT, PbS-TBAI, and ZnO-SCN layers, each followed by its corresponding ligand treatment. The ZnO layer is positioned atop the PbS stack, serving to electrically isolate it from the underlying Ag electrode. The resulting conduction path follows Au-SCN / PbS-EDT / PbS-TBAI / ZnO-SCN / Ag-SCN, supporting both in-plane charge transport within individual layers and vertical carrier movement across the heterojunction interfaces, as schematically depicted in Fig. 8a. Upon illumination with 1480 nm light, the photodiode exhibits an open-circuit voltage of 0.22 V and achieves a remarkable external quantum efficiency of 49%-56.1 % at a -4 V bias. The built-in electric field established across the device (Fig. 8b) promotes efficient charge separation, yielding a response time below 20 ms—a value that exceeds the temporal resolution limit of our

measurement system.

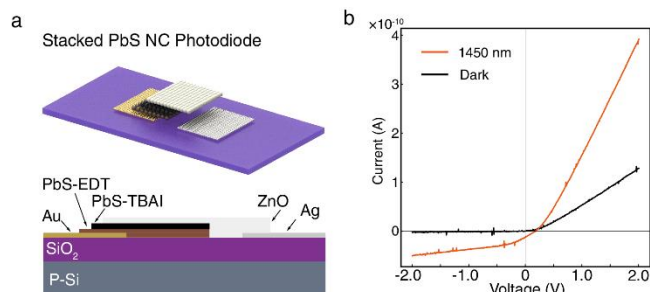


Figure 8. **a** Schematic illustration of the device architecture for all-printed infrared photodiodes based on a PbS NC/ZnO NC heterojunction. **b** I-V characteristics of all-printed 1500 nm PbS NC/ZnO NC IR photodiodes measured in the dark (black) and under 1480 nm illumination (orange), demonstrating photoresponse and rectifying behavior.

6. Conclusion

We report a general approach for the layer-by-layer printing and chemical functionalization of NC structures. In contrast to conventional inkjet printing followed by high-temperature sintering, our method integrates nano-resolution EHDP with room-temperature L-E. This synergistic combination substantially broadens the palette of compatible NC inks, substrate materials, and attainable device functionalities. Crucially, the entirely room-temperature process preserves thermally sensitive materials and facilitates the heterogeneous integration of disparate components on diverse substrate platforms.

Using Ag NCs as a model system, we systematically investigated the printing parameters and elucidated the mechanism underlying L-E-assisted sintering. Our findings indicate that the dynamic coordination between NCs and their surface ligands during the printing process governs subsequent chemical fusion. Extending this understanding, we validated the semiconductive properties of printed PbS NCs and successfully demonstrated fully printed multilayer photodiodes responsive to 1500 nm irradiation. Device performance was further enhanced by integrating asymmetric metal electrodes—fabricated directly via metal NC printing, a strategy that offers greater design flexibility compared to conventional post-lithography metallization.

As hybrid electronics increasingly converge toward integrated sensing, communication, and computing functions within a single device, the chemical-assisted nanoprinting method presented here offers a versatile and practical route for the heterogeneous integration of multi-material, multi-functional devices on both rigid and flexible substrate platforms.

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